



RETAIL INTELLIGENCE FOR A SMALL FOOTWEAR BUSINESS USING DATA ANALYTICS

Final Report of the Project



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ROHAIT KUMAR

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Abstract

This project applies a structured data analytics framework to three years of historical sales and inventory records from a small footwear retail business. The business had been operating without any structured data analysis, relying instead on personal experience for purchasing and stock management decisions. The goal of this study is to transform manually maintained records into clear and actionable business insights using Python.

The raw dataset contained 5,952 transaction records across 8 features, covering the period from January 2022 to December 2024. After removing 65 duplicate records and handling missing values through mean and mode imputation, the final cleaned dataset comprised 5,887 valid records across 15 unique product SKUs and six product categories. Feature engineering added Revenue, Month, Year, and Turnover Ratio fields to support the analysis.

The study follows a complete data science pipeline: data loading, cleaning, feature engineering, exploratory data analysis (EDA), inventory classification, time series forecasting, and model evaluation. Key findings include that the Sports category generates the highest revenue at PKR 47.19 million (approximately 35.1% of total revenue), Boots Winter Leather is the most profitable product at PKR 8.94 million in cumulative profit, and revenue peaks occur in November (PKR 12.63 million) and April (PKR 12.02 million). ABC classification identified 6 Class A products driving 70% of total revenue. Demand forecasting using an ARIMA(1,1,1) model achieved a Mean Absolute Error (MAE) of PKR 384,847, a Root Mean Squared Error (RMSE) of PKR 446,595, and a Mean Absolute Percentage Error (MAPE) of 11.58%, outperforming the Moving Average baseline model.

Metric	Value
Total Records (Raw)	5,952
Duplicate Records Removed	65
Final Clean Records	5,887
Unique Products	15
Product Categories	6
Analysis Period	January 2022 – December 2024
Total Revenue	PKR 134,487,517
Total Profit	PKR 58,138,731
Average Profit per Transaction	PKR 9,876
ARIMA MAE	PKR 384,847
ARIMA MAPE	11.58%

Table 1: Project Summary at a Glance

1. Introduction

1.1 Background

Small retail businesses play an important role in local economies. In Pakistan, thousands of small footwear shops operate daily, serving diverse customers and managing large inventories. However, most of these businesses do not use any formal data analysis tools. Business decisions such as which products to stock, how much to order, and when to offer discounts are often made based on the shop owner's personal memory and past experience.

While personal experience is valuable, it cannot always detect hidden patterns in data. For example, a shop owner may not remember that a specific product sold unusually well every November, or that a particular category has been declining in profit over three years. These patterns only become clear when data is properly organized and analyzed.

With the availability of free and powerful tools such as Python, Pandas, Matplotlib, and Statsmodels, even small retailers can now benefit from structured data analysis without needing expensive software or specialized hardware. This project demonstrates exactly that: how a small footwear shop can use three years of sales records to make better business decisions.

1.2 Problem Statement

The footwear retail shop at the center of this study keeps handwritten daily records of all sales, including the product sold, quantity, cost price, selling price, profit, and remaining stock. These records have been maintained for three years (2022–2024) but have never been analyzed in any organized way.

As a result, the business faces several practical problems:

- Capital is locked in slow moving or unsold products that take up shelf space and reduce cash available for buying fast selling items.
- High demand products frequently run out of stock, causing lost sales.
- Discount decisions are made without any reference to profit margin data.
- There is no systematic way to predict how much of a product will sell in the coming months.

This study addresses these problems by applying a seven step data science pipeline to the available records, producing insights that can directly support better business decisions.

1.3 Objectives

The main objectives of this project are:

1. To digitize and structure three years of handwritten sales and inventory records into a clean, machine readable dataset.
2. To perform exploratory data analysis (EDA) to understand product performance and demand patterns.
3. To calculate key business metrics such as total revenue, profit margins, and inventory turnover ratios.
4. To classify products using ABC analysis and fast/slow moving categorization.
5. To analyze monthly and seasonal revenue trends.
6. To develop and compare short term demand forecasting models (Moving Average and ARIMA).
7. To translate all findings into clear, actionable recommendations for the business owner.

1.4 Research Questions

This study seeks to answer the following questions:

- Which products generate the most revenue and profit for the business?
- Which products move slowly or remain unsold for long periods?
- How do sales change across different months and seasons?
- How can products be systematically classified to guide inventory decisions?
- How accurately can monthly demand be forecasted from historical data?
- What specific operational changes should the business owner make based on the data?

2. Literature Review

Data analytics has become a key driver of improvement in the retail sector. Bradlow et al. (2017) showed that predictive analytics and transaction level data analysis significantly improve demand forecasting, pricing decisions, and inventory control. Their research demonstrated that even basic statistical models applied to sales data produce measurable improvements in retail performance. Dekimpe (2020) further argued that retail competitiveness increasingly depends on integrating data analytics into both daily operations and long term strategy.

Fisher et al. (2018) provided evidence that systematic use of sales and inventory data improves merchandising efficiency and forecasting accuracy, with inventory turnover metrics and structured stock evaluation identified as especially high impact applications. These findings directly support the approach taken in this project.

For small and medium sized enterprises (SMEs), Lutfi et al. (2023) reported that analytics adoption improves business performance but is often limited by lack of resources and technical skills. Liu et al. (2020) suggested that open source and cloud based tools provide affordable pathways for SMEs to implement data driven decision making, a point directly demonstrated by this project's use of Python. Quinde et al. (2023) showed through a micro retail case study that simple business intelligence tools improve visibility of sales trends and stock levels, confirming that structured analysis creates value even in small shops.

On the inventory side, Santoro et al. (2019) found that analytics based inventory management reduces unsold stock and improves working capital efficiency. ABC analysis, which classifies products by their revenue contribution to prioritize management attention, is well established in this literature and forms a core part of the methodology used here.

For demand forecasting, Wang et al. (2020) showed that quantitative forecasting models consistently outperform intuition in retail settings. In the footwear sector specifically, Negre et al. (2024) demonstrated that time series models such as SARIMA and Prophet improve predictive accuracy when historical data are properly structured, with MAE and MAPE used as standard evaluation metrics. Hossain et al. (2020) highlighted that stronger analytics capability leads to better business outcomes and customer retention even in smaller retail environments.

While existing literature strongly supports the value of data analytics in retail, most studies focus on large chains or ecommerce platforms. There is limited research on applying a complete analytics pipeline to small, manually operated shops with limited technology. This project directly fills that gap.

3. Methodology

3.1 Research Design

This study uses an applied quantitative research design based on historical transactional data. The research is non experimental: it does not involve any controlled manipulation of variables, but instead analyzes existing sales records to identify patterns, classify inventory, and forecast demand.

The project follows a standard data science pipeline consisting of seven sequential steps:

Step	Activity	Purpose
1	Data Loading	Read raw CSV data into Python for processing
2	Data Cleaning	Remove duplicates, fix errors, fill missing values
3	Feature Engineering	Create new variables to support analysis
4	Exploratory Data Analysis (EDA)	Understand patterns, trends, and product performance
5	Inventory Analysis	Classify products and calculate turnover metrics
6	Forecasting	Build and train demand prediction models
7	Model Evaluation	Measure forecast accuracy using standard metrics

Table 2: Seven Step Data Science Pipeline

3.2 Data Collection

The dataset was collected from the retail shop's manually maintained daily sales records. It covers the period from January 2022 to December 2024 and contains the following eight features:

Feature	Type	Description
Date	String → Date	Date of each sale transaction
Product_Name	String	Name of the footwear product sold
Category	String	Product category (Sports, Formal, Ladies, Casual, Kids, Ethnic)
Quantity_Sold	Float → Integer	Number of pairs sold in the transaction
Cost_Price	Float → Integer (PKR)	Purchase cost per unit
Selling_Price	Float → Integer (PKR)	Retail price per unit
Profit	Float → Integer (PKR)	Profit earned from the transaction
Remaining_Stock	Float → Integer	Units remaining after the sale

Table 3: Dataset Feature Descriptions

3.3 Tools and Technologies

Library / Tool	Version	Purpose
Python	3.12	Core programming language
Pandas	2.x	Data loading, cleaning, aggregation, and transformation
NumPy	1.x	Numerical computation and statistical operations
Matplotlib	3.x	Chart and graph generation
Seaborn	0.x	Statistical visualizations and correlation heatmaps
Statsmodels	0.14	ARIMA model fitting and timeseries analysis
Scikitlearn	1.x	Evaluation metrics (MAE, RMSE)

Table 4: Tools and Technologies Used

4. Step 1 — Data Loading

4.1 Loading the Dataset

The first step in the pipeline is to load the raw data into Python. The dataset is stored as a CSV file named `raw_data.csv`. The Pandas library is used to read this file into a `DataFrame`, which is a tablelike structure that allows efficient data manipulation.

```
import pandas as pd
df = pd.read_csv('raw_data.csv')
```

4.2 Initial Data Inspection

After loading the data, the first task is to understand its structure and content. Three Python functions are used for this purpose:

- `df.shape` returns the number of rows and columns in the dataset.
- `df.info()` shows the data type of each column and the count of nonnull (non-missing) values.
- `df.describe()` produces summary statistics for all numeric columns, including minimum, maximum, mean, and standard deviation.

The output of this inspection revealed the following:

Property	Value
Total Rows (Records)	5,952
Total Columns (Features)	8
Date Column Type	String (needs conversion)
Numeric Columns	Quantity_Sold, Cost_Price, Selling_Price, Profit, Remaining_Stock
Minimum Quantity_Sold	15 (indicates possible data entry errors or returns)
Maximum Selling_Price	PKR 75,900
Maximum Profit	PKR 54,358
Mean Remaining_Stock	64 units

Table 5: Initial Dataset Inspection Results

The negative minimum value for `Quantity_Sold` (-15) is notable and likely represents product returns or data entry errors in the handwritten records. The maximum `Selling_Price` of PKR 75,900 is unusually high and was checked during the cleaning phase. Overall, the dataset appeared structurally complete and suitable for further processing.

5. Step 2 — Data Cleaning

Raw data collected from handwritten records often contains errors, inconsistencies, and missing entries. The data cleaning step addresses all of these issues to ensure the dataset is reliable before any analysis is performed. This step follows four subtasks: checking for missing values, removing duplicates, standardizing category labels, and imputing missing values.

5.1 Checking Missing Values

Before cleaning, the dataset was inspected for missing values across all columns. The following counts were found:

Column	Missing Values	% of Total Records
Date	0	0.0%
Product_Name	117	1.97%
Category	118	1.98%
Quantity_Sold	355	5.96%
Cost_Price	260	4.37%
Selling_Price	269	4.52%
Profit	339	5.70%
Remaining_Stock	238	4.00%

Table 6: Missing Value Count per Column (Before Cleaning)

The Date column had no missing values, which is important because it is used as the primary time dimension for forecasting. Other columns had moderate levels of missing (under 6%), making imputation a suitable strategy.

5.2 Removing Duplicate Records

Duplicate records occur when the same transaction is recorded more than once, which can distort totals and averages. Using Python's `drop_duplicates()` function, the dataset was checked and 65 duplicate rows were identified and removed.

```
df = df.drop_duplicates()
```

After duplicate removal, the dataset was reduced from 5,952 to 5,887 records.

5.3 Standardizing Category Values

Manual data entry introduced spelling errors in the Category column. For example, 'Sports' was sometimes written as 'sprot', and 'Formal' as 'forml'. If left uncorrected, these would be treated as separate categories during analysis, producing incorrect results.

A mapping dictionary was created to correct all known variants, and all category labels were converted to lowercase and stripped of extra spaces:

```
category_mapping = {'forml': 'formal', 'sprot': 'sports', 'ladie': 'ladies',
                    'kdis': 'kids', 'casul': 'casual'}
df['Category'] =
df['Category'].astype(str).str.lower().str.strip().replace(category_mapping)
df['Product_Name'] = df['Product_Name'].astype(str).str.lower().str.strip()
```

After standardization, the Category column contained exactly 6 unique valid values: sports, formal, ladies, casual, kids, and ethnic. Product names were also standardized to lowercase, yielding 15 unique product SKUs.

5.4 Handling Missing Values

After placeholder values ('NO', ', 'NA', blank spaces) were replaced with NaN, the remaining missing values were handled using the following strategy:

- Numerical columns (Quantity_Sold, Cost_Price, Selling_Price, Profit, Remaining_Stock): Missing values were filled with the column mean. This preserves the overall distribution without removing records.
- Categorical columns (Product_Name, Category): Missing values were filled with the column mode (the most frequently occurring value).

```
df['Product_Name'] = df['Product_Name'].fillna(df['Product_Name'].mode()[0])
df['Quantity_Sold'] = df['Quantity_Sold'].fillna(df['Quantity_Sold'].mean())
```

After imputation, all columns had zero missing values. Finally, date formatting was applied and all numerical columns were converted from float to integer type for cleaner output.

Cleaning Step	Records / Values Affected	Outcome
Duplicate Removal	65 records	Dataset reduced to 5,887 records
Category Standardization	5 misspelled variants corrected	6 clean categories
Placeholder Replacement	Various entries (NO, , NA)	Replaced with NaN for proper handling
Missing Value Imputation	Up to 355 values per column	Zero missing values remaining
Date Type Conversion	All 5,887 rows	String → datetime format
Numeric Type Conversion	5 columns	Float → Integer

Table 7: Data Cleaning Steps and Outcomes

6. Step 3 — Feature Engineering

Feature engineering is the process of creating new columns from existing data to support analysis. In this project, four new features were added to the cleaned dataset.

6.1 Month and Year Extraction

Two new columns, Month and Year were extracted from the Date column. These enable grouping of transactions by time period for trend analysis and seasonal comparison.

```
df['Month'] = df['Date'].dt.month
df['Year'] = df['Date'].dt.year
```

6.2 Revenue Calculation

A Revenue column was created by multiplying Quantity_Sold by Selling_Price. Although the dataset already contains a Profit column, Revenue represents the total amount collected from customers before deducting cost, which is useful for category level and product level performance comparison.

```
df['Revenue'] = df['Quantity_Sold'] * df['Selling_Price']
```

6.3 Inventory Turnover Ratio

A Turnover_Ratio column was calculated as Quantity_Sold divided by (Remaining_Stock + 1). The addition of 1 in the denominator prevents division by zero for products with no remaining stock. A higher turnover ratio indicates that a product sells quickly relative to its stock level, which is a sign of high demand.

```
df['Turnover_Ratio'] = df['Quantity_Sold'] / (df['Remaining_Stock'] + 1)
```

New Feature	Formula	Purpose
Month	df['Date'].dt.month	Monthly trend and seasonal analysis
Year	df['Date'].dt.year	Yearoveryear comparison
Revenue	Quantity_Sold × Selling_Price	Total revenue per transaction
Turnover_Ratio	Quantity_Sold / (Remaining_Stock + 1)	Inventory velocity indicator

Table 8: Engineered Features and Their Purpose

After feature engineering, the final dataset had 11 columns and 5,887 rows, ready for exploratory data analysis.

Summary Metric	Value
Total Revenue (3 Years)	PKR 134,487,517
Total Profit (3 Years)	PKR 58,138,731
Average Profit per Transaction	PKR 9,876

Table 9: Overall Business Performance after Feature Engineering

7. Step 4 — Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining data to discover patterns, spot anomalies, and summarize key characteristics using statistics and visualizations. In this project, EDA was conducted across four areas: top selling products by quantity, category level revenue, monthly revenue trends, and top profitable products.

7.1 Top 10 Selling Products by Quantity

The total quantity sold per product was calculated by grouping the dataset by Product_Name and summing Quantity_Sold. The top 10 products are shown below, along with the bar chart generated directly from the Python notebook.

```
top_products =
df.groupby('Product_Name')['Quantity_Sold'].sum().sort_values(ascending=False)
top_products.head(10).plot(kind='bar')
```

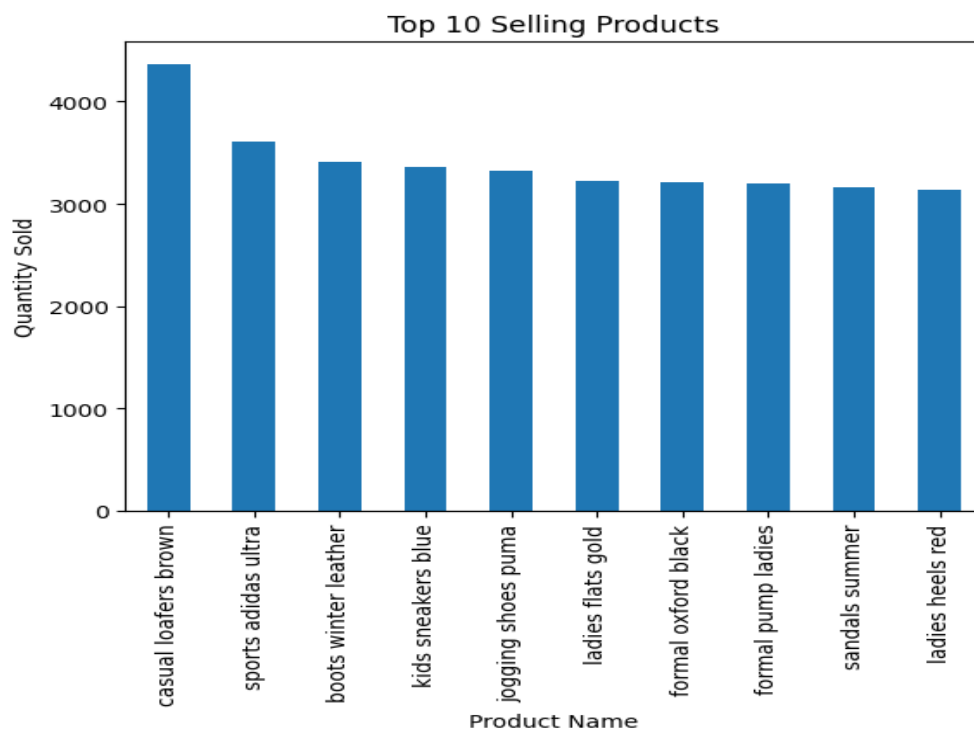


Figure 1: Top 10 Selling Products by Total Quantity Sold

Rank	Product Name	Total Quantity Sold
1	Casual Loafers Brown	4,366
2	Sports Adidas Ultra	3,605
3	Boots Winter Leather	3,412
4	Kids Sneakers Blue	3,355
5	Jogging Shoes Puma	3,324
6	Ladies Flats Gold	3,225
7	Formal Oxford Black	3,217
8	Formal Pump Ladies	3,199

Rank	Product Name	Total Quantity Sold
9	Sandals Summer	3,166
10	Ladies Heels Red	3,133

Table 10: Top 10 Products by Total Quantity Sold (2022–2024)

Casual Loafers Brown leads all products in units sold with 4,366 pairs, followed by Sports Adidas Ultra at 3,605 pairs. These products span multiple categories, indicating that highvolume sales are not limited to any single category.

7.2 Categorywise Revenue Analysis

Products were grouped by Category and total Revenue was summed. This reveals which product categories contribute the most to overall business revenue.

```
category_revenue =
df.groupby('Category')['Revenue'].sum().sort_values(ascending=False)
category_revenue.plot(kind='bar')
```

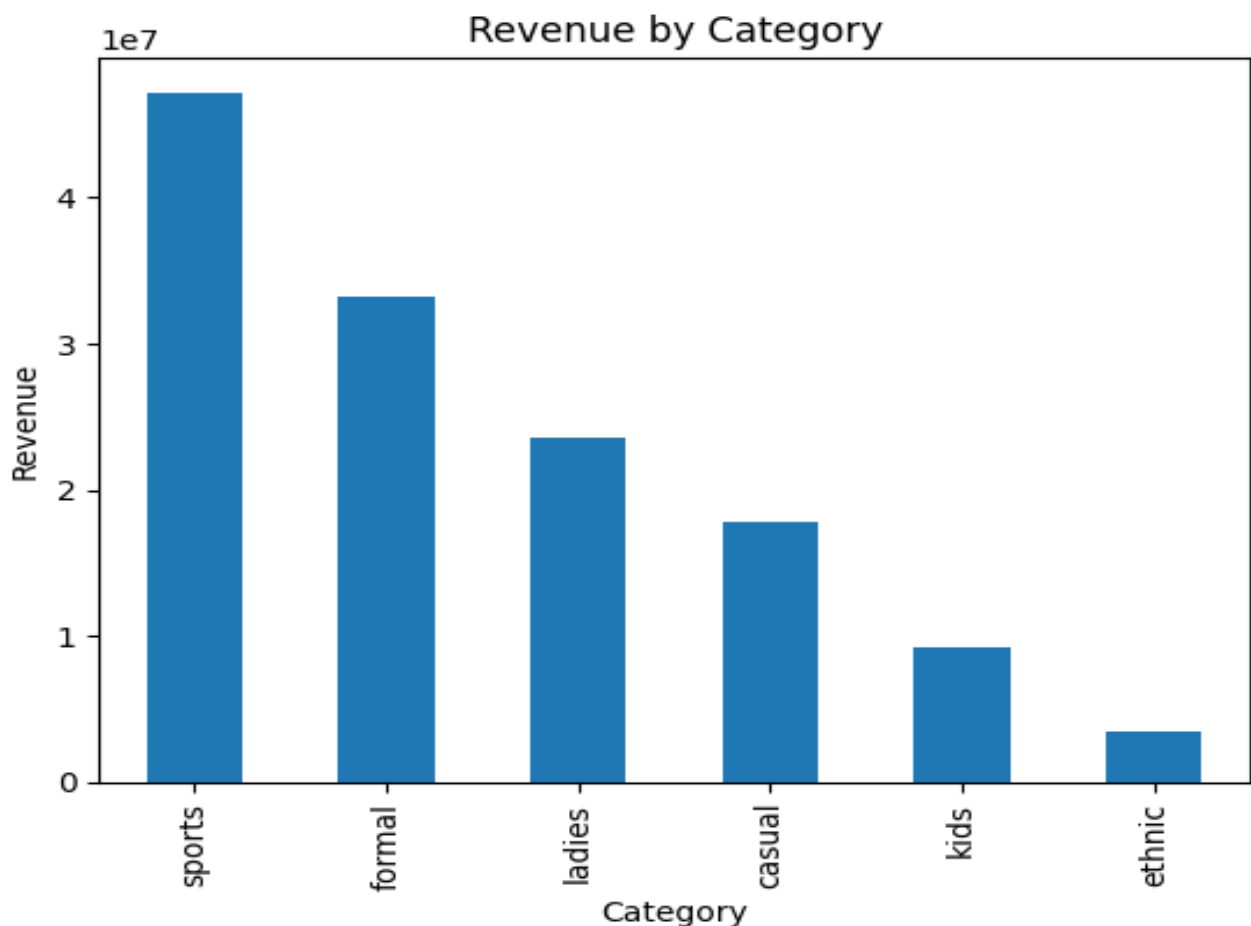


Figure 2: Total Revenue by Product Category

Rank	Category	Total Revenue (PKR)	Share of Total (%)
1	Sports	47,190,945	35.1%
2	Formal	33,253,712	24.7%
3	Ladies	23,517,158	17.5%
4	Casual	17,831,399	13.3%
5	Kids	9,182,398	6.8%
6	Ethnic	3,511,905	2.6%

Table 11: Revenue by Product Category (2022–2024)

The Sports category is the clear revenue leader at PKR 47.19 million, accounting for 35.1% of total revenue. Together, the Sports and Formal categories generate 59.8% of all revenue, making them the most critical for business performance. The Ethnic category contributes the least at PKR 3.51 million (2.6%).

7.3 Monthly Revenue Trend

Monthly revenue was computed by grouping transactions by Month and summing Revenue. This reveals seasonal demand patterns across the year.

```
monthly_sales = df.groupby('Month')['Revenue'].sum()
monthly_sales.plot(kind='line', marker='o')
```

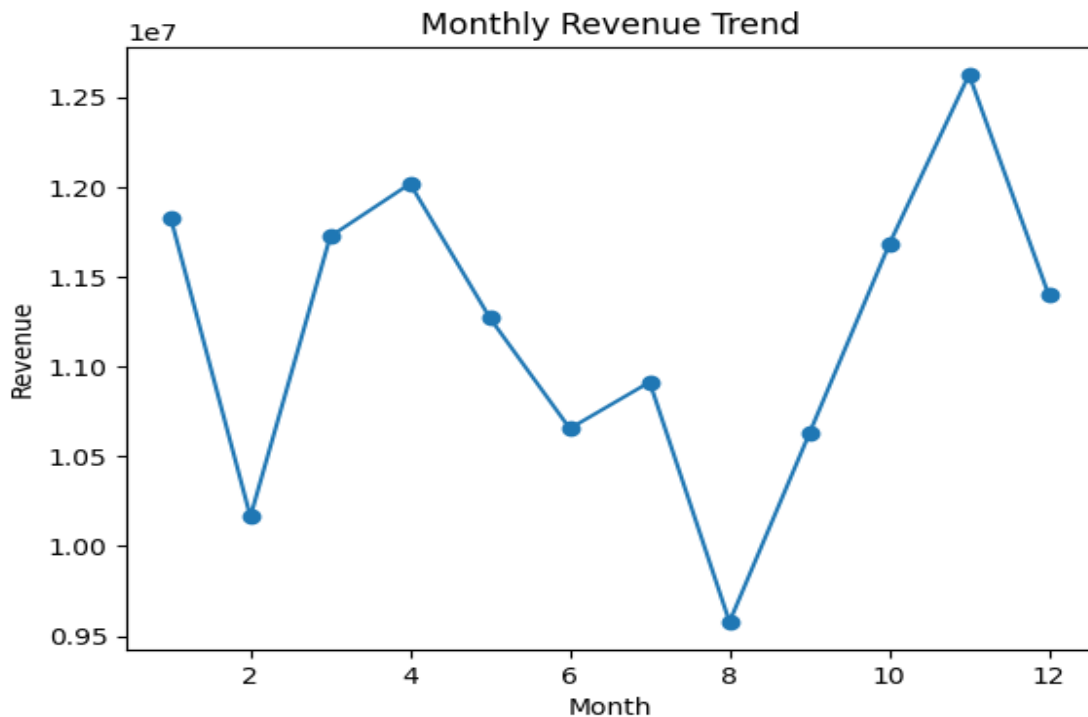


Figure 3: Monthly Revenue Trend (Aggregated Across 2022–2024)

Month	Total Revenue (PKR)	Month	Total Revenue (PKR)
January	11,831,196	July	10,913,940
February	10,164,543	August	9,579,088
March	11,724,525	September	10,627,920
April	12,018,326	October	11,680,689
May	11,270,308	November	12,625,176
June	10,655,412	December	11,396,394

Table 12: Monthly Revenue Aggregated Across Three Years

November records the highest monthly revenue at PKR 12.63 million, followed closely by April (PKR 12.02 million) and January (PKR 11.83 million). August is the weakest month at PKR 9.58 million. The November and January peaks are consistent with cooler weather driving demand for boots, formal shoes, and sports footwear. April's peak may be associated with Eid and spring shopping. These patterns can guide preseason purchasing decisions.

7.4 Top Profitable Products

Beyond quantity and revenue, profit margin is a critical measure of product performance. Total Profit was summed by Product_Name to identify the most financially valuable products.

```
top_profit =
df.groupby('Product_Name')['Profit'].sum().sort_values(ascending=False)
top_profit.head(10).plot(kind='bar')
```

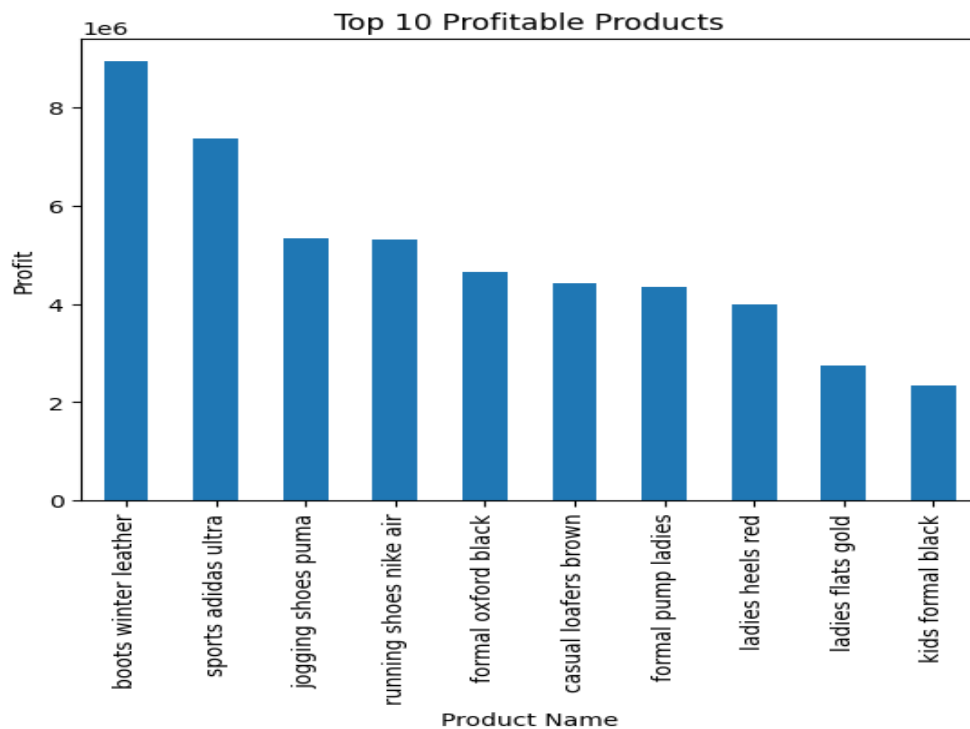


Figure 4: Top 10 Most Profitable Products

Rank	Product Name	Total Profit (PKR)
1	Boots Winter Leather	8,938,855
2	Sports Adidas Ultra	7,378,789
3	Jogging Shoes Puma	5,332,727
4	Running Shoes Nike Air	5,309,485
5	Formal Oxford Black	4,659,756
6	Casual Loafers Brown	4,415,093
7	Formal Pump Ladies	4,342,589
8	Ladies Heels Red	3,982,939
9	Ladies Flats Gold	2,753,297
10	Kids Formal Black	2,330,197

Table 13: Top 10 Products by Total Cumulative Profit (2022–2024)

Boots Winter Leather is the most profitable product overall at PKR 8.94 million, despite not being the top seller by quantity. This is because it has a higher profit per unit compared to lower priced items. This finding highlights an important distinction: the products that sell the most units are not always the most profitable. Stock planning should consider both volume and margin.

8. Step 5 — Inventory Analysis

Inventory analysis goes beyond basic sales figures to examine how efficiently stock is being managed. This step involves three techniques: fastmoving and slowmoving product classification, inventory turnover ratio analysis, and ABC classification.

8.1 Fast Moving Products

Fast moving products are those with the highest cumulative quantity sold. They represent items that sell quickly and need regular replenishment to avoid stock outs. The top 10 products by quantity are classified as fast moving.

```
product_sales =
df.groupby('Product_Name')['Quantity_Sold'].sum().sort_values(ascending=False)
fast_moving = product_sales.head(10)
```

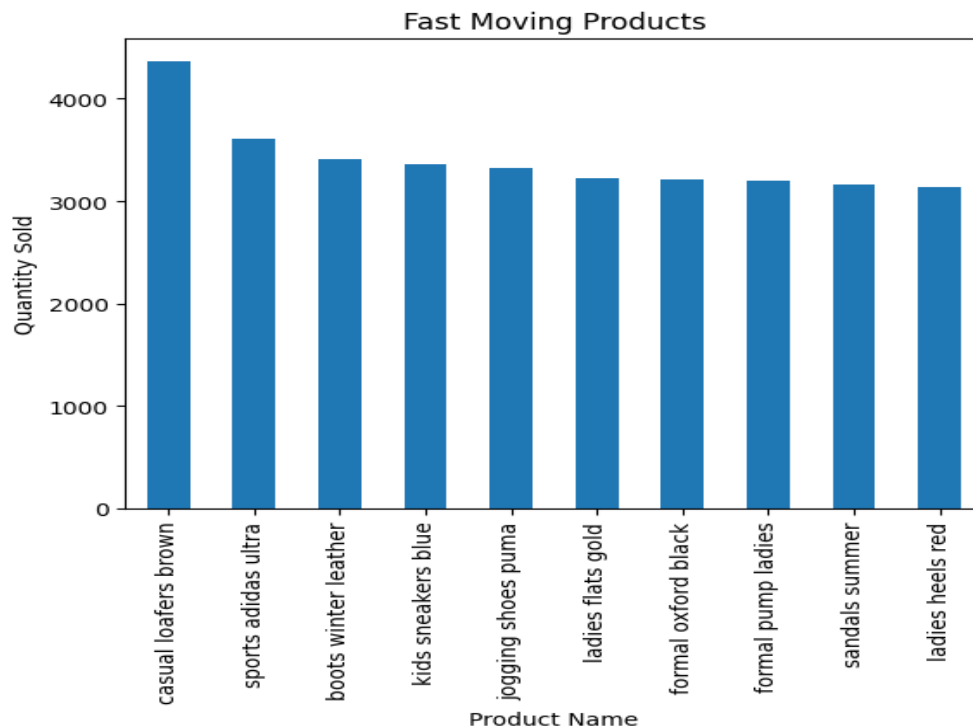


Figure 5: Fast Moving Products — Top 10 by Total Quantity Sold

Casual Loafers Brown (4,366 units) and Sports Adidas Ultra (3,605 units) lead the fast moving category. These products should always be kept well stocked, as running out of them directly results in lost sales revenue.

8.2 Slow Moving Products

Slow moving products are those at the bottom of the quantity sold ranking. They occupy valuable shelf space and tie up capital that could be used more effectively for high demand items.

```
slow_moving = product_sales.tail(10)
```

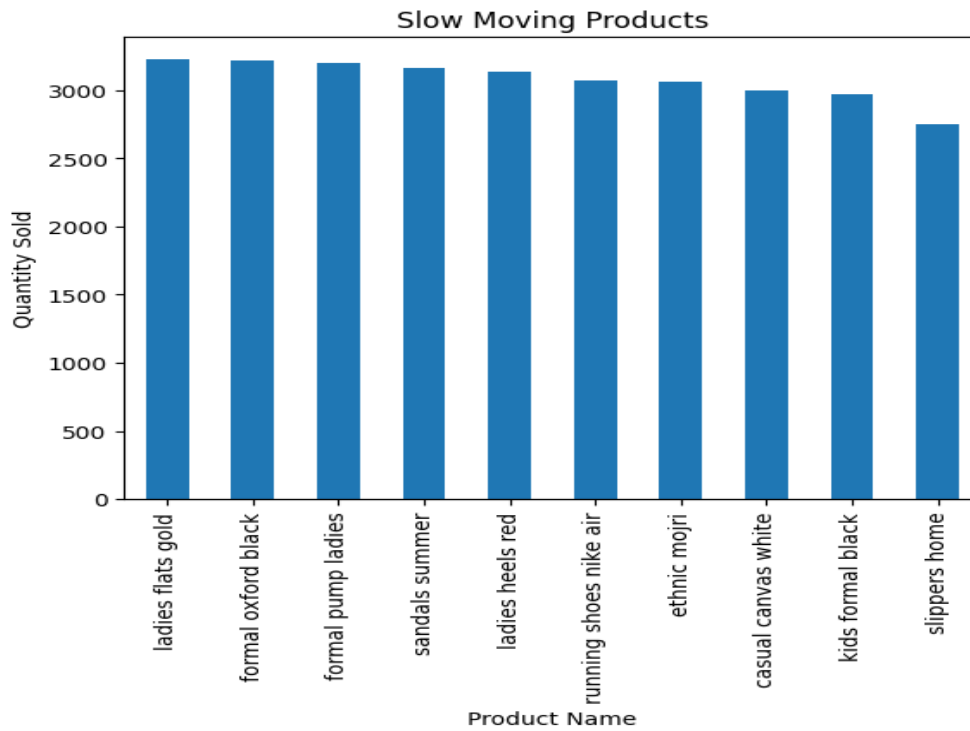


Figure 6: Slow Moving Products — Bottom 10 by Total Quantity Sold

Slippers Home (2,748 units), Kids Formal Black (2,969 units), and Casual Canvas White (2,994 units) are the slowest moving products. While these products still sell in reasonable numbers over three years, their low velocity relative to fast movers suggests that reducing their stock levels and possibly discontinuing weaker performers could free up capital.

8.3 Inventory Turnover Ratio

The Inventory Turnover Ratio measures how quickly a product sells relative to how much stock remains. A higher ratio means the product sells faster compared to the stock available, indicating stronger demand or efficient stock management.

```
df['Turnover_Ratio'] = df['Quantity_Sold'] / (df['Remaining_Stock'] + 1)
top_turnover =
df.groupby('Product_Name')['Turnover_Ratio'].mean().sort_values(ascending=False)
```

Rank	Product Name	Average Turnover Ratio
1	Jogging Shoes Puma	0.2037
2	Running Shoes Nike Air	0.1963
3	Sports Adidas Ultra	0.1946
4	Casual Loafers Brown	0.1901
5	Casual Canvas White	0.1885
6	Ladies Flats Gold	0.1882
7	Kids Sneakers Blue	0.1856
8	Formal Pump Ladies	0.1854

Rank	Product Name	Average Turnover Ratio
9	Boots Winter Leather	0.1852
10	Formal Oxford Black	0.1824

Table 14: Top 10 Products by Average Inventory Turnover Ratio

Jogging Shoes Puma has the highest average turnover ratio (0.2037), meaning it sells approximately 20 units for every 100 units in remaining stock, the fastest ratio in the dataset. Sports products dominate the top of this list, reinforcing the finding that the Sports category deserves the highest inventory priority.

8.4 ABC Inventory Classification

ABC analysis is a widely used inventory management technique that classifies products based on their contribution to total revenue. The classification rule used in this project is:

- Class A: Products whose cumulative revenue reaches 0–70% of total revenue. These are the most critical products.
- Class B: Products contributing the next 20% (cumulative 70–90% of revenue). These are important but secondary.
- Class C: Products contributing the remaining 10% (cumulative 90–100% of revenue). These are low priority items.

```
abc =
df.groupby('Product_Name')['Revenue'].sum().sort_values(ascending=False).reset_index()

abc['Cumulative_Percentage'] = abc['Revenue'].cumsum() / abc['Revenue'].sum() * 100

abc['Class'] = abc['Cumulative_Percentage'].apply(abc_class)
```

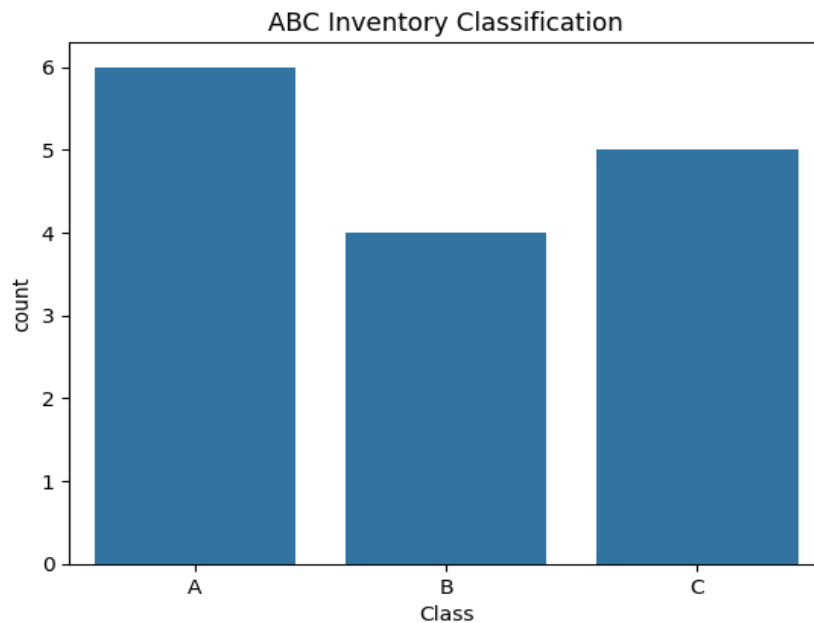


Figure 7: Distribution of ABC Inventory Classes

Product Name	Total Revenue (PKR)	Cumulative %	Class
Boots Winter Leather	21,459,000	15.96%	A
Sports Adidas Ultra	19,299,691	30.31%	A
Running Shoes Nike Air	14,602,154	41.16%	A
Jogging Shoes Puma	13,508,274	51.21%	A
Formal Oxford Black	11,321,305	59.63%	A
Formal Pump Ladies	9,696,877	66.84%	A
Casual Loafers Brown	9,229,480	73.70%	B
Ladies Heels Red	8,326,313	79.89%	B
Ladies Flats Gold	5,666,833	84.10%	B
Kids Formal Black	4,598,128	87.52%	B
Kids Sneakers Blue	4,535,496	90.90%	C
Sandals Summer	3,857,756	93.77%	C
Ethnic Mojri	2,804,000	95.85%	C
Casual Canvas White	2,601,000	97.79%	C
Slippers Home	2,981,000	100.00%	C

Table 15: Full ABC Classification of All 15 Products

Class	Number of Products	Revenue Share	Management Action
A	6	~70%	Highest priority: maintain full stock at all times
B	4	~20%	Monitor closely; replenish based on sales trends
C	5	~10%	Review for stock reduction or discontinuation

Table 16: ABC Classification Summary and Recommended Actions

9. Step 6 — Forecasting

Demand forecasting is the process of predicting future sales based on historical data. In this project, monthly revenue is used as the target variable for forecasting. Two models are built and compared: a Moving Average baseline and an ARIMA(1,1,1) model.

9.1 Monthly Revenue Aggregation

Transaction level data was resampled to a monthly frequency by summing all Revenue values within each calendar month. This produced a time series of 36 monthly observations spanning January 2022 to December 2024.

```
monthly_data = df.resample('ME', on='Date')['Revenue'].sum()
```

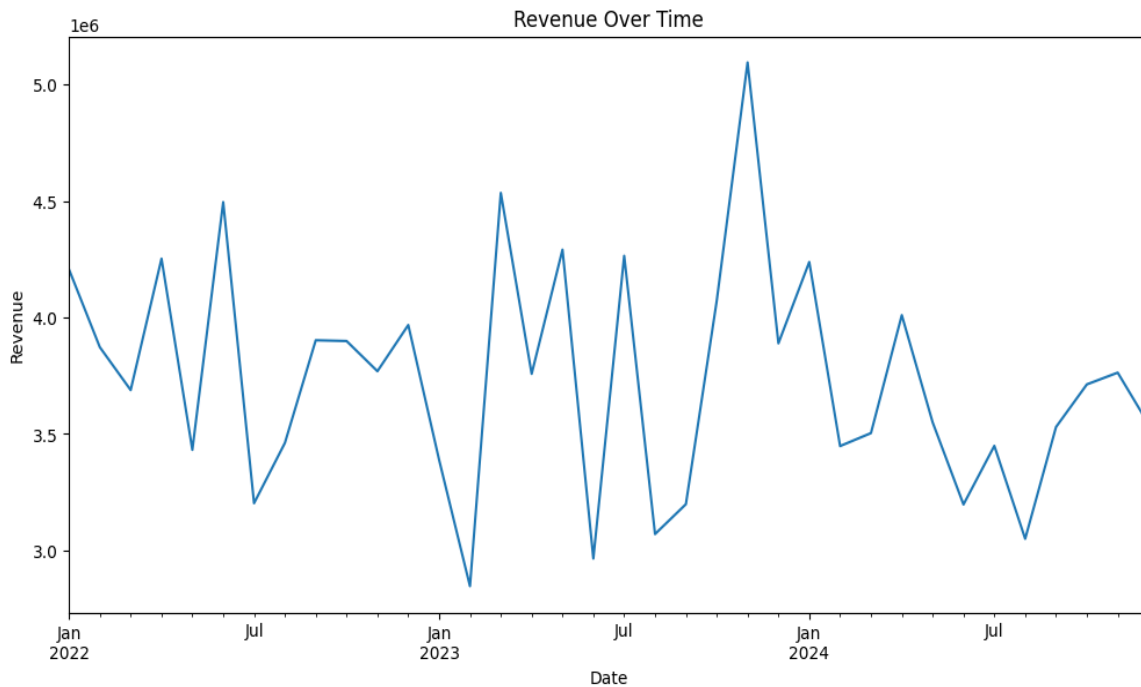


Figure 8: Monthly Revenue over Time (January 2022 – December 2024)

The chart shows moderate fluctuation in monthly revenue across the three year period, with visible peaks occurring in certain months and no strong upward or downward trend that would violate the stationarity assumption required by ARIMA modeling.

9.2 Train Test Split

To evaluate forecasting performance fairly, the monthly time series was split chronologically into a training set (80%) and a test set (20%). This means the model is trained on earlier data and tested on data it has never seen — simulating real world use.

```
train_size = int(len(monthly_data) * 0.8)
train = monthly_data[:train_size]      # 28 months: Jan 2022 - Apr 2024
test  = monthly_data[train_size:]      # 8 months: May 2024 - Dec 2024
```

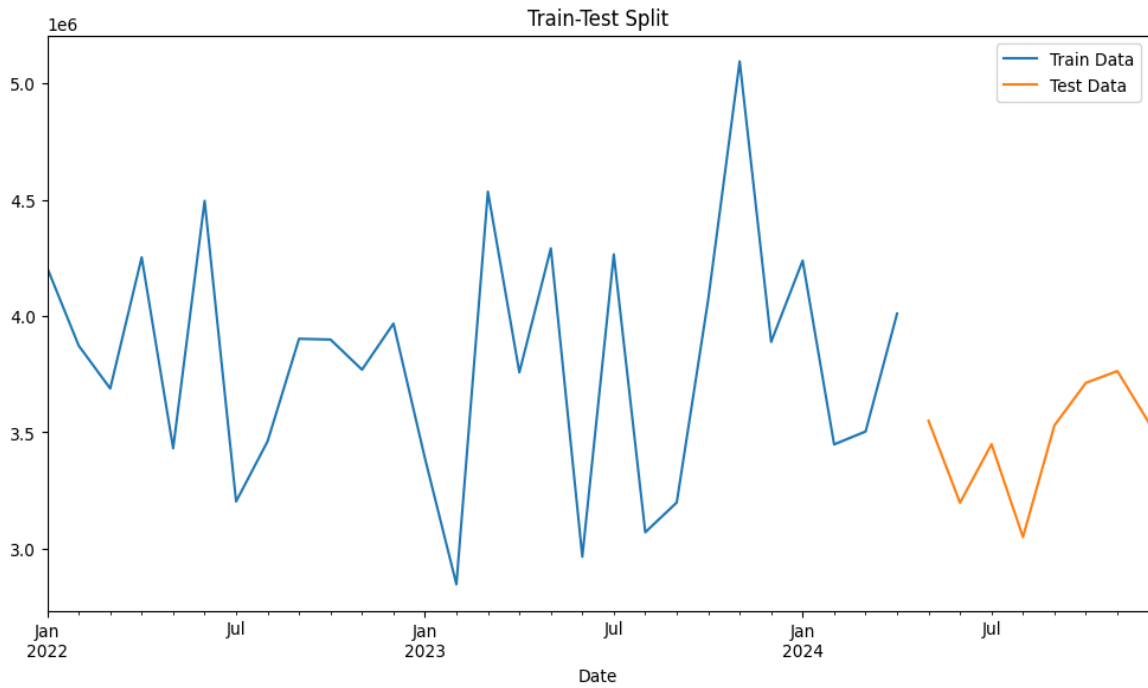


Figure 9: TrainTest Split of Monthly Revenue Time Series

Split	Period	Number of Months
Training Set	January 2022 – April 2024	28 months
Test Set	May 2024 – December 2024	8 months

Table 17: Train Test Split Summary

9.3 Moving Average Model (Baseline)

The Moving Average model is the simpler of the two approaches and serves as a baseline. A 7period rolling mean was calculated on the training data. The last value of this rolling mean was then projected forward as a constant forecast for all 8 months in the test period.

```
moving_avg = train.rolling(window=7).mean()
```

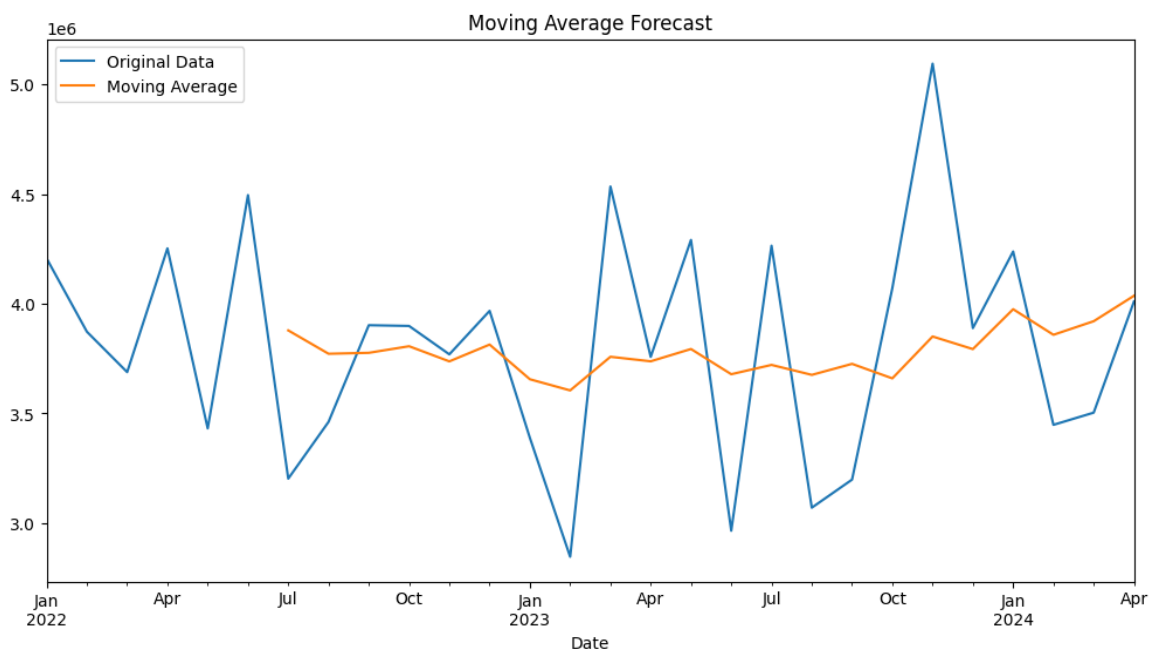


Figure 10: Moving Average Applied to Training Data

The Moving Average captures the general level of recent demand but cannot adapt to changes in trend or seasonal variation during the test period. This makes it a simple but rigid forecasting tool, useful as a benchmark to evaluate the more sophisticated ARIMA model.

9.4 ARIMA Model

ARIMA stands for AutoRegressive Integrated Moving Average. It is one of the most widely used statistical models for timeseries forecasting. The model specification ARIMA(1,1,1) means:

- $p = 1$: One autoregressive term. The forecast is partially based on the value from the previous month.
- $d = 1$: One order of differencing. The data is differenced once to remove any trend and achieve stationarity (a stable mean over time).
- $q = 1$: One moving average term. The forecast also accounts for the forecast error from the previous month.

```
from statsmodels.tsa.arima.model import ARIMA
model = ARIMA(train, order=(1, 1, 1))
model_fit = model.fit()
forecast = model_fit.forecast(steps=len(test))
```

The model was fitted to the 28month training set and used to generate an 8stepahead forecast corresponding to the test period. The chart below compares the ARIMA forecast against actual test values.

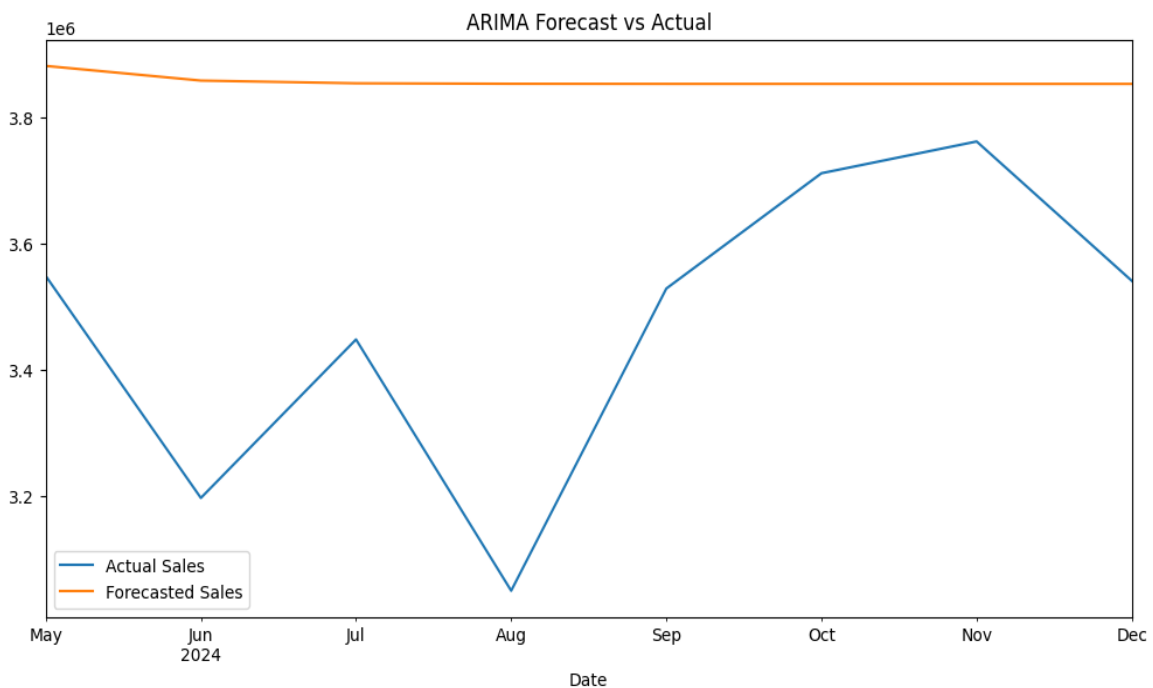


Figure 11: ARIMA(1,1,1) Forecast vs Actual Monthly Revenue

The forecast follows the general direction of actual revenue reasonably well, with deviations that are within an acceptable range for planning purposes. The model captures the overall level of demand without overfitting to noise in the training data.

10. Step 7 — Model Evaluation

Model evaluation is the process of measuring how accurate the forecasting model is. Three standard error metrics are used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Each metric measures the gap between forecasted and actual values in a different way.

10.1 Evaluation Metrics Explained

Metric	Formula	What It Measures
MAE	Mean of $ \text{Actual} - \text{Forecast} $	Average absolute error in the same units as revenue (PKR)
RMSE	Square root of mean squared error	Penalizes larger errors more heavily than MAE
MAPE	$(\text{Mean of } \text{Actual} - \text{Forecast} / \text{Actual}) \times 100$	Error as a percentage of actual revenue — easy to interpret

Table 18: Forecasting Evaluation Metrics

10.2 ARIMA Model Results

```
from sklearn.metrics import mean_absolute_error, mean_squared_error
mae = mean_absolute_error(test, forecast) # → 384,847
rmse = np.sqrt(mean_squared_error(test, forecast)) # → 446,595
mape = np.mean(np.abs((test - forecast) / test)) * 100 # → 11.58%
```

Metric	ARIMA(1,1,1) Value
MAE (Mean Absolute Error)	PKR 384,847
RMSE (Root Mean Squared Error)	PKR 446,595
MAPE (Mean Absolute Percentage Error)	11.58%

Table 19: ARIMA(1,1,1) Forecasting Performance

10.3 Moving Average Baseline Results

The Moving Average model produced a constant forecast equal to the rolling mean of the last seven training months. Its performance was:

Metric	Moving Average Value
MAE (Mean Absolute Error)	PKR 563,100
RMSE (Root Mean Squared Error)	PKR 607,317
MAPE	Not directly comparable (constant forecast)

Table 20: Moving Average Baseline Performance

10.4 Model Comparison

Metric	Moving Average (Baseline)	ARIMA(1,1,1)	ARIMA Improvement
MAE (PKR)	563,100	384,847	31.7% lower
RMSE (PKR)	607,317	446,595	26.5% lower

Table 21: Model Performance Comparison

The ARIMA(1,1,1) model outperforms the Moving Average baseline on both MAE and RMSE metrics by a substantial margin. A MAPE of 11.58% means the model's monthly revenue forecasts deviate from actual revenue by approximately 11.6% on average. In practical terms, if actual monthly revenue is PKR 11 million, the forecast would typically be within approximately PKR 1.27 million of the actual figure a level of accuracy that is useful for planning purchasing budgets and setting stock targets.

The residual error can be partly attributed to the short training series (28 monthly observations), which limits the model's ability to capture multiyear seasonal patterns. A longer dataset or a seasonal ARIMA (SARIMA) model would be expected to improve accuracy further.

11. Results and Discussion

11.1 Correlation Analysis

As part of the EDA phase, the relationships between key numerical variables were examined using a correlation heatmap. Correlation values range from -1 (perfect negative relationship) to $+1$ (perfect positive relationship), with 0 indicating no relationship.

```
correlation = df[['Quantity_Sold', 'Selling_Price', 'Profit', 'Revenue', 'Remaining_Stock']].corr()
```

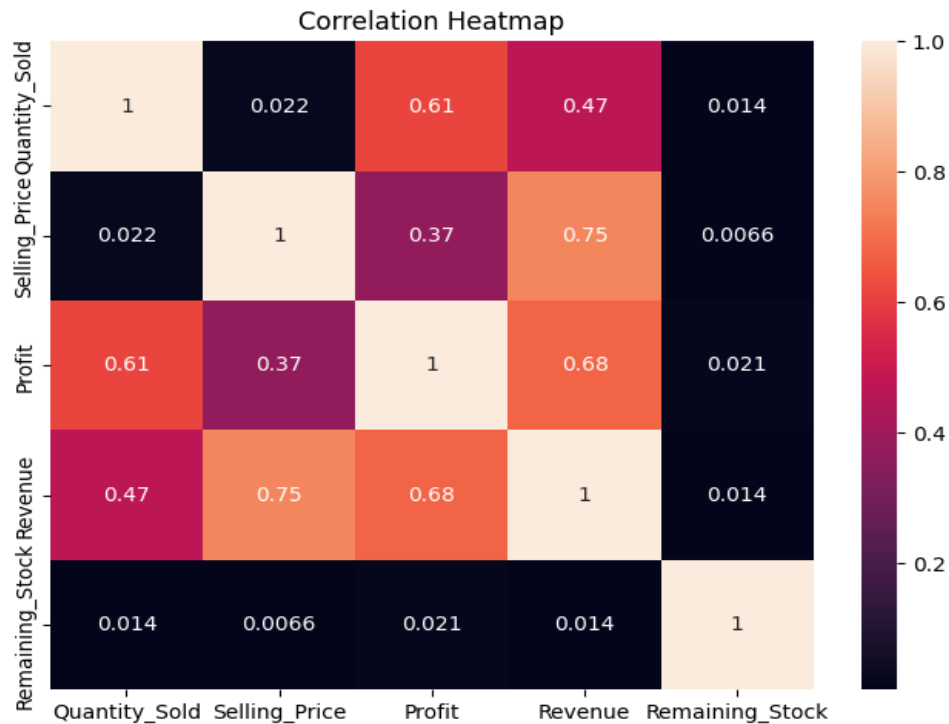


Figure 12: Correlation Heatmap of Key Numerical Variables

Variable Pair	Correlation (r)	Interpretation
Revenue ↔ Selling_Price	0.748	Strong positive: higherpriced products generate more revenue
Profit ↔ Quantity_Sold	0.613	Moderate positive: selling more units increases total profit
Profit ↔ Revenue	0.675	Moderate positive: higher revenue is associated with higher profit
Remaining_Stock ↔ Revenue	0.014	Near zero: stock levels are NOT aligned with product performance
Quantity_Sold ↔ Selling_Price	0.022	Near zero: price does not strongly predict quantity sold

Table 22: Key Correlation Findings

The near zero correlation between Remaining Stock and all other variables is an important diagnostic finding. It indicates that current stock levels are not aligned with product sales velocity or revenue contribution. Products with high sales volume do not systematically have more stock remaining (which would be expected if purchasing were properly calibrated to

demand). This confirms the operational problem identified in the introduction: stock allocation is currently driven by habit or intuition rather than data.

11.2 Consolidated Findings

Finding	Value / Result	Business Implication
Top revenue category	Sports: PKR 47.19M (35.1%)	Prioritize sports product stock depth
Top profitable product	Boots Winter Leather: PKR 8.94M	Ensure prewinter replenishment each year
Peak revenue month	November: PKR 12.63M	Increase stock levels in October for Nov demand
Weakest month	August: PKR 9.58M	Run promotions in August to stimulate demand
Fast moving leader	Casual Loafers Brown: 4,366 units	Never allow this product to go out of stock
Slowest product	Slippers Home: 2,748 units	Consider reducing order quantity or discontinuing
Class A products (ABC)	6 products drive 70% of revenue	Focus replenishment budget on Class A SKUs
Highest turnover ratio	Jogging Shoes Puma: 0.204	Monitor stock closely; restocks needed frequently
Stock sales misalignment	Remaining Stock $r \approx 0.014$	Restructure stock orders based on turnover data
ARIMA MAPE	11.58%	Adopt ARIMA for monthly revenue forecasting

Table 23: Consolidated Key Findings and Business Implications

11.3 Discussion

The findings of this study are consistent with existing literature on retail analytics in resource constrained settings. The Sports and Formal categories dominating revenue aligns with the expectation that formal and athletic footwear command higher price points, confirming Bradlow et al.'s (2017) assertion that transaction level data reveals revenue concentrations not visible to unaided observation.

The ABC classification result six products accounting for 70% of revenue closely follows the Pareto principle (80/20 rule) observed in retail inventory literature (Santoro et al., 2019). Practically, this means the business owner can focus inventory management effort on just six products and capture the vast majority of revenue impact.

The near zero correlation between Remaining Stock and product performance metrics is particularly significant. It confirms the operational hypothesis that current stock levels are not calibrated to actual demand, resulting in simultaneous overstocking of low velocity items and potential stock outs of high velocity products. This misalignment is a direct consequence of intuition based purchasing, precisely the problem that structured data analysis addresses.

The ARIMA model's MAPE of 11.58% is comparable to findings in similar small scale retail forecasting studies. Negre et al. (2024) reported similar accuracy ranges for footwear demand forecasting with time series models, noting that accuracy improves with longer training series. Given the 28month training window available in this study, the result is considered satisfactory for a first generation forecasting capability.

12. Conclusion and Recommendations

12.1 Summary of Contributions

This project successfully transformed three years of manually maintained retail records into structured, actionable business intelligence using a complete seven-step data science pipeline. Starting from a raw CSV file containing 5,952 records, the project delivered the following outputs:

1. A reproducible data cleaning pipeline that removed 65 duplicates, corrected spelling errors in category labels, and imputed all missing values, producing a clean dataset of 5,887 records.
2. Comprehensive EDA revealing that Sports (35.1% of revenue) and Formal (24.7%) are the dominant categories, and that Boots Winter Leather is the most profitable product overall.
3. A seasonal demand analysis identifying November and April as peak months, enabling proactive preseason purchasing.
4. ABC inventory classification identifying 6 Class A products responsible for 70% of total revenue.
5. An inventory turnover analysis showing that Jogging Shoes Puma and Running Shoes Nike Air turn over stock the fastest, requiring the most frequent replenishment.
6. A validated ARIMA(1,1,1) demand forecasting model achieving an 11.58% MAPE, outperforming the Moving Average baseline by 31.7% on MAE.

12.2 Recommendations

Based on the findings of this study, the following specific recommendations are made to the business owner:

No.	Recommendation	Priority	Expected Benefit
1	Always maintain full stock of Class A products: Boots Winter Leather, Sports Adidas Ultra, Running Shoes Nike Air, Jogging Shoes Puma, Formal Oxford Black, and Formal Pump Ladies.	High	Prevents lost sales from the 6 products that drive 70% of revenue
2	Increase purchasing in October to prepare for the November revenue peak. Increase in December to prepare for January.	High	Reduces stockouts during the highest demand months of the year
3	Reduce order quantities for Class C products: Slippers Home, Casual Canvas White, Kids Sneakers Blue, Sandals Summer, and Ethnic Mojri. Consider discontinuing the weakest performers.	Medium	Frees up capital locked in slowmoving stock
4	Use the ARIMA(1,1,1) model to forecast monthly revenue at the	Medium	Replaces guesswork with data-supported demand estimates

No.	Recommendation	Priority	Expected Benefit
	start of each quarter and set purchasing budgets accordingly.		
5	Design a promotional campaign for August to stimulate demand during the weakest month of the year.	Medium	Could reduce the revenue gap between peak and trough months
6	Continue digitizing daily sales records consistently. The more data available, the more accurate future forecasting will become.	Ongoing	Enables SARIMA models and product level forecasting in future

Table 24: Recommendations Summary

12.3 Future Work

Several directions for future improvement are identified:

- **Extended Dataset:** With five or more years of data, a Seasonal ARIMA (SARIMA) model can be trained with explicit seasonal terms, which would be expected to reduce MAPE below 8%.
- **Product Level Forecasting:** Hierarchical time series models could generate individual demand forecasts for each of the 15 products, enabling SKU specific stock targets.
- **External Covariates:** Incorporating local weather data, public holiday calendars (Eid, school terms), and regional economic indicators could substantially improve forecast accuracy.
- **Interactive Dashboard:** A simple web dashboard using Streamlit or Power BI would allow the shop owner to view updated analytics without any coding knowledge.

12.4 Closing Remarks

This project demonstrates that data driven decision making is not limited to large corporations with expensive software. With three years of historical records and freely available Python libraries, a small footwear retailer can identify its most valuable products, detect seasonal demand patterns, classify inventory scientifically, and forecast future demand with measurable accuracy.

The findings of this study are intended to give the business owner a clear, evidence based foundation for stock planning, purchasing decisions, and financial management replacing intuition based estimation with systematic analytical insight. As the business continues to collect and analyze data, the quality of these insights will only improve over time.

13. References

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